

Perfect hindsight would solve up to 16 more tasks in 100 than the best single view. None of 8 learned choices beat always using the cheapest view on both success and cost.

THE AGENT LOOP, AND WHERE THE DECISION SITS



Fig 1. Only the page view changes. The dashed box is what we measure: one fixed view, perfect hindsight, or a learned choice, each judged on **both** success and cost.

RESULTS ACROSS 8,934 TASK ATTEMPTS

+16.35 in 100

PERFECT HINDSIGHT · SUCCESS ONLY · VS ONE FIXED VIEW

0 of 8

LEARNED CHOICES THAT WON ON SUCCESS AND COST

1 of 8

HINDSIGHT CHOICES THAT WON ON SUCCESS AND COST

ONE FIXED VIEW — the highest-success view in a setting, used for every task.

ALWAYS-CHEAPEST — the lowest-average-cost view in a setting, used for every task. “Won” = beat it on both success and cost.

PERFECT HINDSIGHT — after each task, select a view that solved it. Optimistic: rerunning one unchanged view flips 10–14% of outcomes and gains 2.0–7.6 tasks in 100 by itself.

THE CATCH

The agents that gain most from choosing a view solve the fewest tasks — so they produce the fewest examples to learn from.

WHY LEARNING THE CHOICE FAILS HERE

A “which view?” label exists only after a solved task. The best fixed view solves just 2–36%, leaving 15–97 usable labels per setting. Only 2 of 6 VisualWebArena settings have enough to train a classifier.

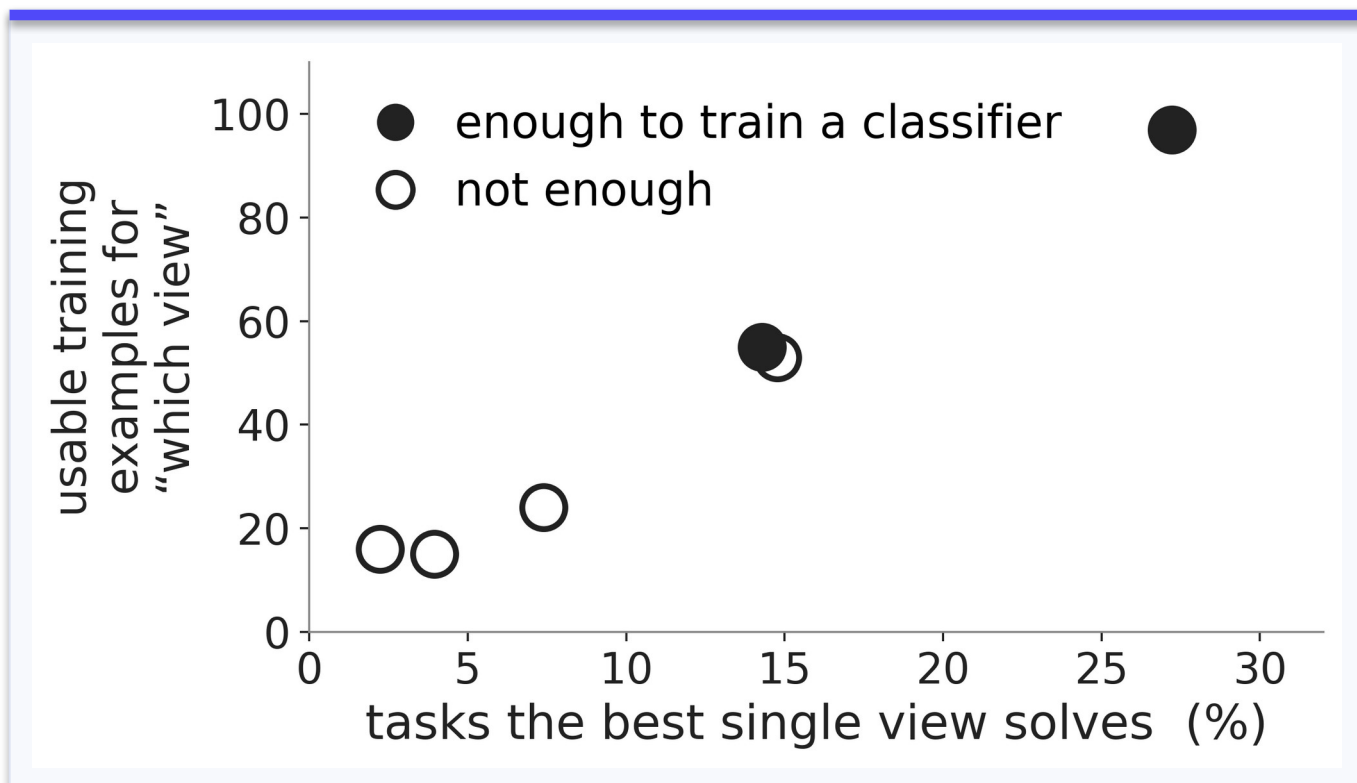


Fig 3. Usable “which view?” labels against best-fixed-view success, one point per VisualWebArena setting.

Deliberately shrinking the training data points to scarcity: failing settings need at least 2.1–4.2× more tasks than the benchmarks contain.

TAKEAWAY

Learning how to see is not only a modelling problem. It depends on how good the agent producing the examples already is. Improve the agent, collect reliable examples, then learn when to look.

Measured in the 2–36% success regime we observed. This need not hold for stronger agents.

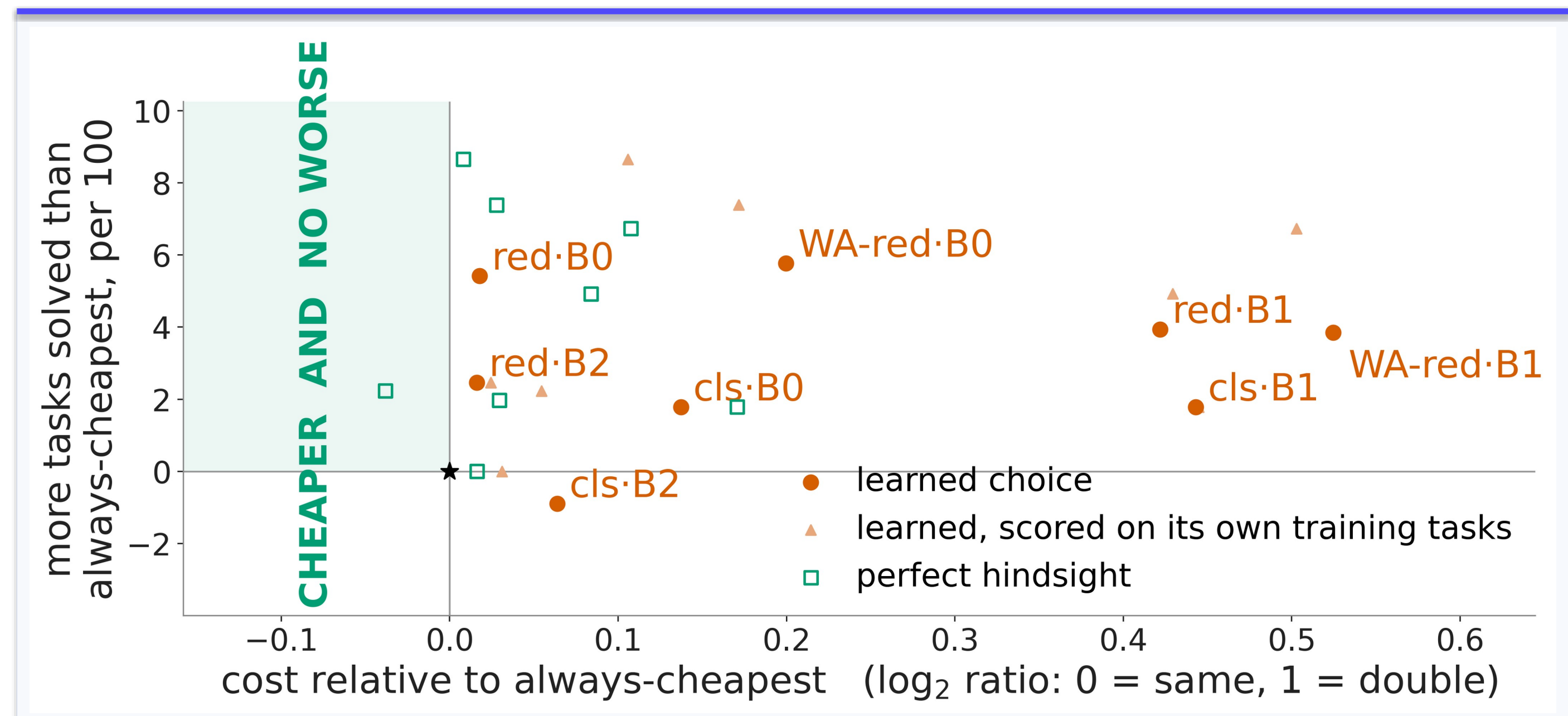


Fig 2. Every choice is compared with **always use the cheapest view** (★). A win lands in the shaded region: cheaper and no worse. Cheapest means lowest average cost, so some points sit left of ★. Learned choices use held-out tasks.

Perfect hindsight would pay. After-the-fact selection solves 3.45–16.35 more tasks in 100 than the best fixed view and spends 1.6–35.3% less, in all 8 settings.

Nothing we trained could do it. In 0 of 8 settings did a learned choice beat always-cheapest on both success and cost. Hindsight does so in only 1 of 8.

What survives is a bound, not a method. Sending never-solved tasks to the cheapest view saves 9.5–30.6% at equal success in 8 of 8, but needs hindsight; plain always-cheapest usually saves more.

WATCH THE LAPTOP BESIDE THIS POSTER

Same task, three views, different behaviour and bills — step by step. Each view wins one illustrative task; this is **not** how often each wins.

	LOOK	READ	BOTH
“Navigate to the item on this page whose image is taken during a sunset.”	✓ 2 steps · \$0.007	✗ 21 steps · \$0.096	✓ 3 steps · \$0.014
“Navigate to my listing of the blue bike and change the price to \$85.50 (including in the description).”	✗ 30 steps · \$0.116	✓ 9 steps · \$0.079	✗ 6 steps · \$0.029
“Show me the cheapest bike with red handlebars between \$900–950.”	✗ 19 steps · \$0.075	✗ 6 steps · \$0.025	✓ 7 steps · \$0.037

One recorded attempt per view, B0 · classifieds. ✓ / ✗ repeated independently; steps and bill vary between runs.

